

Akshay Thakur

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Education

Ph.D., Aerospace and Mechanical Engineering GPA: 3.968 / 4.0

Aug 2022 – Present

University of Notre Dame, Indiana, USA

M.S., Mechanical Engineering GPA: 3.96 / 4.0

Aug 2022 – Aug 2023

University of Notre Dame, Indiana, USA

Research Experience

Research Intern

Jun 2026 – Oct 2026

Synopsys Inc., USA

- Development of foundation models for external aerodynamics.
- Benchmarking of existing foundation models and development of new foundation models.
- Uncertainty quantification for foundation model predictions.
- Design of data acquisition and training strategies for large-scale aerodynamic datasets.

Research Intern

Sep 2025 – Feb 2026

Mitsubishi Electric Research Labs, Cambridge, Massachusetts, USA

- Development and evaluation of physics-grounded, scalable surrogate models for sound propagation based on attention-based neural operators and graph neural networks.

Graduate Research Assistant

Aug 2023 – Present

CAMO Lab, Aerospace and Mechanical Engineering Dept., University of Notre Dame, Indiana, USA

- Development of physics-informed neural network-based methodologies for boundary layer and shock-based problems.
- Development and evaluation of novel finite element-based numerical simulation methods for PDEs by incorporating deep learning and reduced-order modeling techniques for enhanced computational efficiency.
- Developed and validated neural network-based surrogate models for Riemann solvers used in the modeling and simulation of hyperbolic PDEs.
- Contributed to the development of an in-house finite element package written in Julia.
- Developed an unstructured finite volume method-based simulation code in PyTorch.

Graduate Research Assistant

Aug 2022 – Aug 2023

COMSAIL Lab, Aerospace and Mechanical Engineering Dept., University of Notre Dame, Indiana, USA

- Developed and evaluated uncertainty quantification methods for surrogate models based on the Fourier neural operator and its extended variants.

Research Intern

Jul 2021 – Aug 2022

CSCCM Lab, Dept. of Applied Mechanics, Indian Institute of Technology (IIT), Delhi, India

- Developed and implemented a multi-fidelity wavelet-based neural operator for time-independent and time-dependent reliability analysis and design of systems governed by PDEs under uncertainty.
- Developed and implemented a surrogate model for stochastic differential equations using conditional generative moment matching networks.
- Developed and implemented an image-to-image regression capsule network-based framework as a surrogate for PDEs.
- Designed and implemented deep learning models using TensorFlow and PyTorch.

Technical and Research Skills

[GitHub Profile](#) ↗

Languages & Tools: Python, Julia, MATLAB, C++, TensorFlow, PyTorch, CUDA, JAX, MPI

Simulation Software: Ansys Fluent, OpenFOAM

Research Areas: Scientific machine learning, artificial intelligence, model order reduction, finite element and finite volume methods, scientific computing, partial differential equations, uncertainty quantification

Journal Articles

- [1] **Thakur, A.** and Zahr, M.J. (2027). Feature tracking in physics-informed neural networks via joint optimization of analytical nonlinear manifolds: Application to boundary layers. *Computer Physics Communications*, 330, p. 110419. [10.1016/j.cpc.2026.110419](https://doi.org/10.1016/j.cpc.2026.110419).
- [2] **Thakur, A.**, Kumar, S., Zahr, M.J., and Chakraborty, S. (2025). Scalable h-adaptive probabilistic solver for time-independent and time-dependent systems. *arXiv:2508.09623*. *Computer Methods in Applied Mechanics and Engineering*, accepted.
- [3] **Thakur, A.** and Zahr, M.J. (2025). Neural network-based Godunov corrections for approximate Riemann solvers using bi-fidelity learning. *Communications in Applied Mathematics and Computational Science*, Vol. 20, No. 1, pp. 207–229.
- [4] **Thakur, A.** and Chakraborty, S. (2024). MD-NOMAD: Mixture density nonlinear manifold decoder for emulating stochastic simulators and one-shot uncertainty propagation. *Journal of Engineering Mechanics*. [10.1061/JENMDT/EMENG-8613](https://doi.org/10.1061/JENMDT/EMENG-8613).
- [5] Tripura, T., **Thakur, A.**, and Chakraborty, S. (2024). Multi-fidelity wavelet neural operator surrogate model for time-independent and time-dependent reliability analysis. *Probabilistic Engineering Mechanics*, 77, p. 103672.
- [6] **Thakur, A.** and Chakraborty, S. (2023). Deep capsule encoder–decoder network for surrogate modeling and uncertainty quantification. *International Journal for Numerical Methods in Engineering*, 124(12), pp. 2783–2800.
- [7] **Thakur, A.** and Chakraborty, S. (2022). A deep learning based surrogate model for stochastic simulators. *Probabilistic Engineering Mechanics*, 68, p. 103248.

Submitted and In Preparation

- [8] **Thakur, A.** and Zahr, M.J. Feature tracking in physics-informed neural networks via joint optimization of nonlinear deformation manifolds: Application to shocks. *Computer Methods in Applied Mechanics and Engineering*, submitted.
- [9] **Thakur, A.** and Zahr, M.J. Optimization-based nonlinear discontinuous Galerkin method for advection–diffusion problems with high Péclet number. In preparation.

Master's Thesis

- [10] **Thakur, A.** (2023). *Uncertainty Quantification for Signal-to-Signal Regression-Based Neural Operator Frameworks*. University of Notre Dame. doi.org/10.7274/xp68kd2150r.

Conference Proceedings and Presentations

- [11] **Thakur, A.** and Zahr, M. (2024). Exploring Riemann solvers and machine learning surrogates in hyperbolic PDE systems: A neural network-based approach. *16th World Congress on Computational Mechanics and 4th Pan American Congress on Computational Mechanics*, Vancouver, Canada.
- [12] **Thakur, A.** and Kumar, P. (2021). Collision of steam bubbles in a vertical rectangular channel containing subcooled liquid. *Proceedings of the 26th National and 4th International ISHMT-ASTFE Heat and Mass Transfer Conference*, IIT Madras, Chennai, India. Begel House Inc.

Teaching

University of Notre Dame, Indiana

Co-Instructor & Co-Curriculum Developer

Aug 2024 – Dec 2024

- AME 30251: Computing, Numerical Methods, and Data Science in Engineering

Teaching Assistant

Aug 2022 – Present

- AME 30314: Differential Equations, Vibrations, and Control I
- AME 20216: AME Laboratory I
- AME 60635: Intermediate Fluid Dynamics
- Orbital and Space Dynamics

Awards**Outstanding Graduate Student Teaching Award**

Fall 2024

Kaneb Center for Teaching Excellence and The Graduate School, University of Notre Dame, IN

Selected Graduate-Level Courses

University of Notre Dame, Indiana

• Nonlinear and Stochastic Optimization • Advanced Scientific Computing • Statistical Computational Methods • Finite Element Methods • Computational Fluid Dynamics • Advanced Numerical Methods • Mathematical and Scientific Machine Learning

References

Prof. Matthew J. Zahr

Associate Professor, University of Notre Dame, Notre Dame, IN 46556

mzahr@nd.edu | *Ph.D. research advisor*

Prof. Souvik Chakraborty

Associate Professor, Indian Institute of Technology Delhi, New Delhi, India

souvik@am.iitd.ac.in | *Former research advisor*